Impact of Baseline Hard Drug Use on HAART Treatment Response: A Frequentist and Bayesian Analysis of the MACS Cohort

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Introduction

Data for this project were obtained from the Multicenter AIDS Cohort Study (MACS), a prospective cohort study of HIV-1 infection in homosexual and bisexual men followed annually across four U.S. cities after HAART initiation. Year 0 represents each participant's last untreated visit immediately prior to starting HAART and all subsequent visits are on-treatment. The primary exposure is self-reported hard drug use (such as heroin or cocaine) at baseline. Treatment response at Year 2 was evaluated using four outcomes: viral load (log10 copies/mL), CD4+ T-cell count (cells/uL), physical quality of life score (AGG_PHYS), and mental quality of life score (AGG_MENT), both on the SF-36 scale (0–100).

The scientific hypothesis is that baseline hard drug use is associated with poorer treatment response two years after HAART initiation. To evaluate this, for each outcome we compared Year 2 responses between baseline hard drug users and non-users while adjusting for the corresponding baseline outcome value and baseline covariates. For each outcome, the primary statistical hypotheses were:

H_0: There is no difference in Year 2 outcome between baseline hard drug users and non-users ($\beta_1 = 0$),

H_1: Baseline hard drug use is associated with the Year 2 outcome ($\beta_1 \neq 0$)

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where β_1 is the adjusted coefficient for hard drug use in an ANCOVA model for the Year 2 outcome. A secondary analysis adjusts for Year 2 medication adherence to assess whether any observed difference is explained by differential adherence between groups.

Methods

Data Management: The analytic population included HIV-positive participants with complete baseline visit and a Year 2 follow-up visit, and who had ever initiated antiretroviral therapy (everART = 1). Viral load (VLOAD) was log10-transformed to reduce skew and stabilize variance. BMI values coded as 999 or -1 as well as biologically implausible values were set to missing. Race was re-categorized into White Non Hispanic and Other, education into high school or less, some college, and college or higher, smoking status into current versus not current. Year 2 adherence (ADH) was dichotomized as $ $95% versus <95%. The primary exposure hard drug use and all covariates (age, BMI, race, education, smoking status) were fixed at baseline. Data were restructured to wide format pairing Year 0 and Year 2 values per participant. Participants missing the exposure, any covariate, or any outcome were excluded from the analytic dataset via complete-case analysis.

Descriptive Statistics: Baseline characteristics of participants were summarized by baseline hard drug use status. Mean outcome trajectories from Year 0 to Year 2 with 95% confidence bands were plotted by group with individual trajectories for a subsample of 60 participants.

Frequentist and Bayesian Approach: For each of the four outcomes, an ANCOVA framework was used: $Y_2 = \beta_0 + \beta_1(\text{Hard Drug Use}_{baseline}) + \beta_2 Y_0 + \mathbf{X}\beta + \varepsilon$, where Y_2 = Year 2 outcome, Y_0 = Baseline value of the outcome, X = baseline covariates (age, BMI,

race, education, and smoking status). β_1 represents the adjusted difference in Year 2 outcome between baseline hard drug users and non-users, and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. This approach estimated the adjusted mean difference in Year 2 outcomes between hard drug users and non-users, conditional on pre-treatment values and potential confounders. A second set of models was fit adding Year 2 adherence ($\geq 95\%$ vs. $< 95\%$) as a covariate. Comparing the hard drug use coefficient between the unadjusted and adherence-adjusted models allowed assessment of whether differential treatment adherence between groups explained any observed association between baseline drug use and treatment response. The hard drug coefficient was tested at $\alpha = 0.05$ with no multiple comparisons correction. Point estimates are reported with 95% confidence intervals.

Each of the eight ANCOVA models was also fit using Bayesian linear regression implemented in Stan via the cmdstanr interface. The model structure was identical to the frequentist models. Weakly informative priors were placed on all regression coefficients: $\beta \sim Normal(0, 100^2)$. A Half-Normal(0, 100) prior was specified for σ. Posterior distributions were estimated using the Hamiltonian Monte Carlo (HMC) algorithm and its adaptive variant, the No-U-Turn Sampler (NUTS), with four independent chains, each run for 1,000 warm-up and 2,000 sampling iterations. Convergence was confirmed (all R-hat < 1.01; bulk/tail ESS > 400; Appendix Table 5), with satisfactory mixing across chains (Appendix Figures 3–6). For each outcome, the posterior distribution of the hard drug use coefficient was summarized by posterior mean and 95% highest posterior density interval (HPDI). The posterior probability that the coefficient was negative indicating worse outcomes for hard drug users was computed as $\mathrm{P}(\beta < 0)$. To address clinical relevance beyond statistical signif-

icance, the posterior probability that the effect exceeded a pre-specified clinically meaningful threshold was computed as $P(|\beta| > \text{threshold})$, with thresholds of 0.5 log10 copies/mL for viral load, 50 cells/uL for CD4+ T-cell count, and 2 SF-36 points for the QOL outcomes. Model fit between the unadjusted and adherence-adjusted models was compared using Leave-One-Out Information Criterion (LOO-IC). Lower LOO-IC values indicate better predictive accuracy.

All analyses were conducted in R version 4.3.1 (2023-06-16 ucrt), with Bayesian models implemented in Stan via the cmdstanr interface.

Results

Descriptive statistics (Table 1): The analytic cohort included 463 participants, 36 (7.8%) reported baseline hard drug use. The baseline characteristics were largely similar between groups. Hard drug users were more likely to be current smokers (75.0% vs 34.9%) and had slightly lower BMI, while baseline laboratory and QOL measures were comparable. At Year 2, the majority of participants reported $\geq$ 95% adherence. Trajectory plots (Figure 1) showed expected treatment response after HAART initiation: declining viral load and increasing CD4+ T-cell counts in both groups, with stable QOL scores over two years.

Hard Drug Use Effect: Table 2 summarizes frequentist and Bayesian estimates. After adjustment, baseline hard drug use was most strongly associated with lower CD4 count at Year 2 (posterior mean -153 cells/uL; 95% HPDI: -214, -95), with 100% posterior probability of both a negative and clinically meaningful (>50 cells/uL) effect, indicating strong evidence of impaired immune recovery. Physical QOL was also lower (-3.48 points; 95% HPDI: -6.22, -0.79), with 99.5% probability of a negative effect and 86% probability of exceeding the

2-point clinical threshold. In contrast, viral load (-0.01 log10; 95% HPDI: -0.41, 0.41) and mental QOL (-0.53; 95% HPDI: -4.01, 2.99) showed little evidence of clinically meaningful differences, with low probabilities of exceeding predefined thresholds. Estimates were similar with and without adherence adjustment, and frequentist and Bayesian results were closely aligned. Figure 2 shows posterior mass for CD4 and physical QOL lies predominantly below zero, whereas viral load and mental QOL show substantial overlap with the null.

Adherence Effect: Table 3 presents the effect of medication adherence ($\geq$ 95% vs. $<$ 95%) on Year 2 outcomes from the adherence-adjusted models. Higher adherence ($ 95 $ = -0.44, 95% CI: -0.79, -0.09; p = 0.014), with the Bayesian estimate closely agreeing (posterior mean = -0.44, 95% HPDI: -0.78, -0.08). Higher adherence was also associated with significantly greater CD4+ count (β = 55.54, 95% CI: 0.49, 110.58; p = 0.048; Bayesian mean = 54.51, 95% HPDI: 1.92, 106.29). Associations with physical and mental QOL were positive but non-significant (β = 2.13, 95% CI: -0.26, 4.52; p = 0.081 for physical QOL and β = 2.30, p = 0.139 for mental QOL). The posterior probability of a clinically meaningful adherence effect exceeded the pre-specified thresholds for viral load and CD4 outcomes, supporting the clinical relevance of adherence.

Model Fit: Table 4 presents LOO-IC comparisons between unadjusted and adherence-adjusted Bayesian models. The adherence-adjusted model showed modestly better fit for viral load (ΔLOO-IC = -3.2), CD4 count (ΔLOO-IC = -2.6), and physical QOL (ΔLOO-IC = -1.7). For mental QOL, the unadjusted model performed marginally better (ΔLOO-IC = +0.2). However, none of the differences exceeded the commonly used threshold of four LOO-IC units.

Discussion

This analysis examined whether baseline hard drug use was associated with HAART treatment response at Year 2 across laboratory and quality-of-life outcomes in HIV-positive men from the MACS cohort. The strongest association was observed for CD4 count and persisted after adherence adjustment, suggesting immune recovery differences are not fully explained by adherence. Associations with viral load and quality of life outcomes were smaller and less precisely estimated. Medication adherence emerged as an independent predictor of improved laboratory outcomes, reinforcing its central role in HIV treatment and suggesting that adherence support may partially offset the adverse effects of substance use. Frequentist and Bayesian analyses yielded consistent results across all models. The Bayesian framework additionally quantified the probability of clinically meaningful effects and this information unavailable from p-values alone. LOO-IC comparisons showed modest improvements with adherence adjustment for laboratory outcomes, with negligible gain for QOL outcomes.

Limitations include a small hard drug user group (n = 39) limiting precision, potential selection bias from complete-case analysis, and measurement of drug use only at baseline which may not reflect continued or changing use over follow-up. Despite these limitations, the findings highlight the importance of addressing substance use and adherence concurrently in HIV care to optimize immune recovery following HAART initiation.

Table 1: Baseline characteristics (and selected Year 2 follow-up measures) of participants by baseline hard drug use status.

	No	Yes	Overall
	(N=427)	(N=36)	(N=463)
Age (years)			
Mean (SD)	43.1 (8.68)	43.9 (9.52)	43.2 (8.74)
Median [Min, Max]	43.0 [20.0, 73.0]	45.5 [29.0, 61.0]	43.0 [20.0, 73.0]
BMI (kg/m2)			
Mean (SD)	25.3 (4.39)	23.6 (3.45)	25.2 (4.34)
Median [Min, Max]	24.8 [16.5, 45.3]	23.3 [18.0, 31.2]	24.6 [16.5, 45.3]
Race			
White Non-Hispanic	277 (64.9%)	19 (52.8%)	296 (63.9%)
Other	150 (35.1%)	17 (47.2%)	167 (36.1%)
Education			
High school or less	84 (19.7%)	13 (36.1%)	97 (21.0%)
Some college	152 (35.6%)	13 (36.1%)	165 (35.6%)
College+	191 (44.7%)	10 (27.8%)	201 (43.4%)
Smoking status			
Current	149 (34.9%)	27 (75.0%)	176 (38.0%)
Not current	278 (65.1%)	9 (25.0%)	287 (62.0%)
Viral load (log10 copies/mL)			
Mean (SD)	4.53 (0.930)	4.58 (0.871)	4.53 (0.925)
Median [Min, Max]	4.52 [0.237, 8.28]	4.47 [2.87, 6.40]	4.51 [0.237, 8.28]
CD4+ T-cell count			
Mean (SD)	374 (202)	360 (201)	373 (202)
Median [Min, Max]	360 [10.9, 1220]	397 [10.9, 650]	360 [10.9, 1220]
Physical QOL (SF-36, 0-100)			
Mean (SD)	51.4 (9.00)	49.2 (7.05)	51.2 (8.88)
Median [Min, Max]	53.7 [19.2, 69.0]	47.4 [31.4, 62.9]	53.5 [19.2, 69.0]
Mental QOL (SF-36, 0-100)			
Mean (SD)	45.2 (13.8)	42.0 (11.6)	44.9 (13.6)
Median [Min, Max]	49.5 [7.23, 66.0]	44.4 [22.5, 59.6]	49.1 [7.23, 66.0]
Adherence at Year 2			
<95%	45 (10.5%)	1 (2.8%)	46 (9.9%)
>=95%	382 (89.5%)	35 (97.2%)	417 (90.1%)

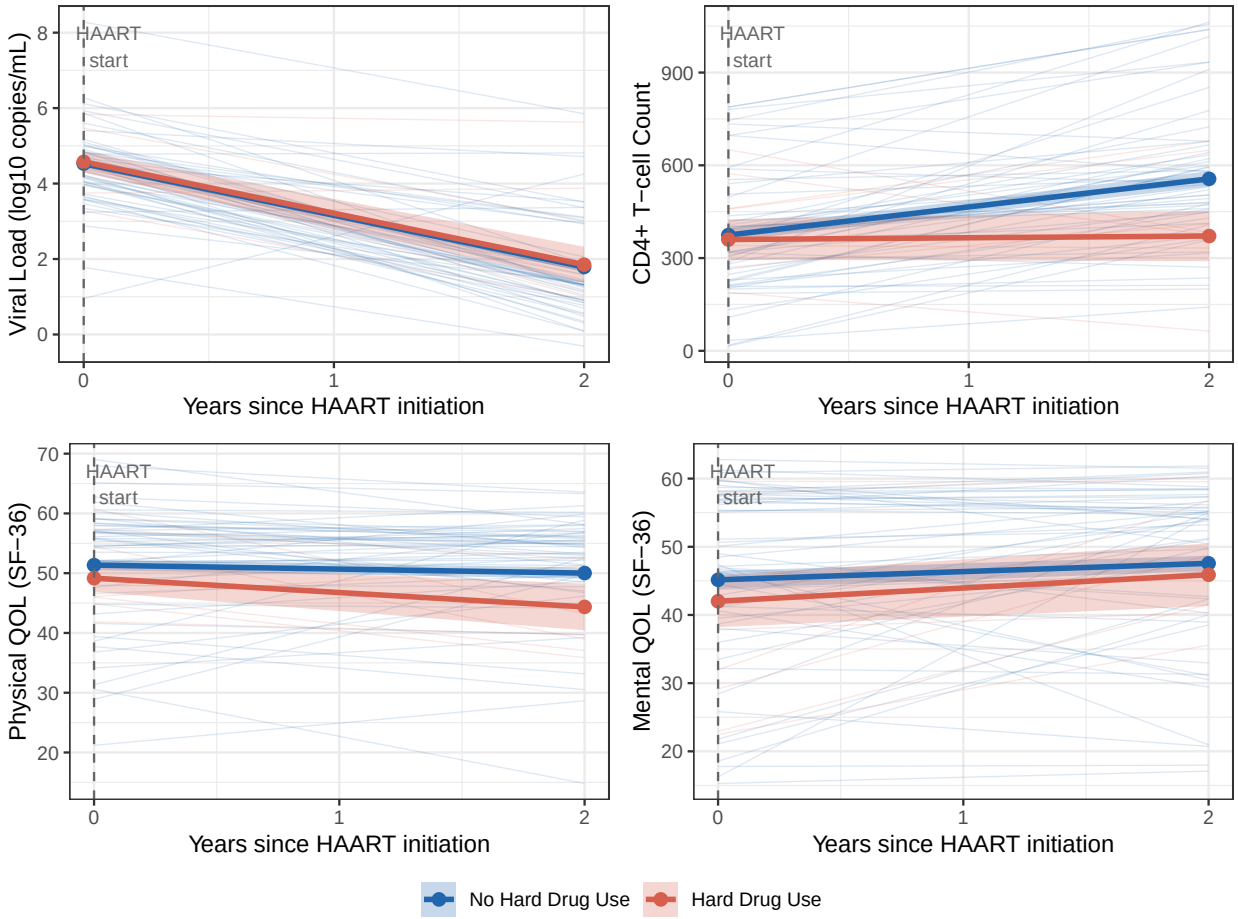

Figure 1: Mean outcome trajectories from Year 0 to Year 2 by baseline hard drug use status. Bold lines show group means with 95 percent confidence bands; faded lines show individual trajectories for a random subsample of 60 participants.

Table 2: Effect of baseline hard drug use on Year 2 outcomes, unadjusted and adjusted for adherence.

		Frequentist		Bayesian		
Outcome	Model	Est (95% CI)	p	Mean (95% HPDI)	P(b<0)	P(Clin.)
CD4 Count	Unadjusted	-163.97 (-226.54, -101.41)	<0.001	-146.49 (-206.82, -88.18)	1.000	0.999
	Adjusted	-169.86 (-232.50, -107.23)	<0.001	-152.79 (-214.23, -94.84)	1.000	1.000
Mental QOL	Unadjusted	-0.28 (-3.74, 3.19)	0.876	-0.26 (-3.75, 3.19)	0.557	0.265
	Adjusted	-0.52 (-4.00, 2.96)	0.769	-0.53 (-4.01, 2.99)	0.612	0.294
Physical QOL	Unadjusted	-3.25 (-5.97, -0.53)	0.019	-3.26 (-5.95, -0.52)	0.993	0.816
	Adjusted	-3.48 (-6.20, -0.75)	0.012	-3.48 (-6.22, -0.79)	0.995	0.859
Viral Load (log10)	Unadjusted	-0.06 (-0.46, 0.34)	0.761	-0.06 (-0.44, 0.32)	0.621	0.015
	Adjusted	-0.01 (-0.41, 0.38)	0.942	-0.01 (-0.41, 0.41)	0.528	0.019

Note:
Clin. thresholds: Viral load 0.5 log10 copies/mL; CD4 50 cells/uL; QOL 2 points.
* P(b<0) = posterior probability effect is negative.
† P(Clin.) = posterior probability effect exceeds clinical threshold.

Table 3: Effect of adherence (95 percent or above vs below 95 percent) on Year 2 outcomes. Thresholds: 0.5 log10 copies/mL (Viral load), 50 cells/uL (CD4), 2 points (QOL).

Outcome	Freq: Est (95% CI)	Freq: p	Bayes: Mean (95% HPDI)	Bayes: P(beta<0)	Bayes: P(Clin)
Viral Load (log10)	-0.44 (-0.79, -0.09)	0.014	-0.44 (-0.78, -0.08)	0.993	0.362
CD4 Count	55.54 (0.49, 110.58)	0.048	54.51 (1.92, 106.29)	0.021	0.569
Physical QOL	2.13 (-0.26, 4.52)	0.081	2.13 (-0.10, 4.58)	0.039	0.543
Mental QOL	2.30 (-0.75, 5.34)	0.139	2.32 (-0.59, 5.39)	0.063	0.586

Table 4: LOO-IC comparison: adherence-adjusted vs unadjusted models. Lower LOO-IC = better fit.

Outcome	LOO-IC (No adherence)	LOO-IC (With adherence)	Delta LOO-IC
Viral Load (log10)	1441.6	1438.4	-3.2
CD4 Count	6126.2	6123.6	-2.6
Physical QOL	3221.9	3220.2	-1.7
Mental QOL	3446.1	3446.3	0.2

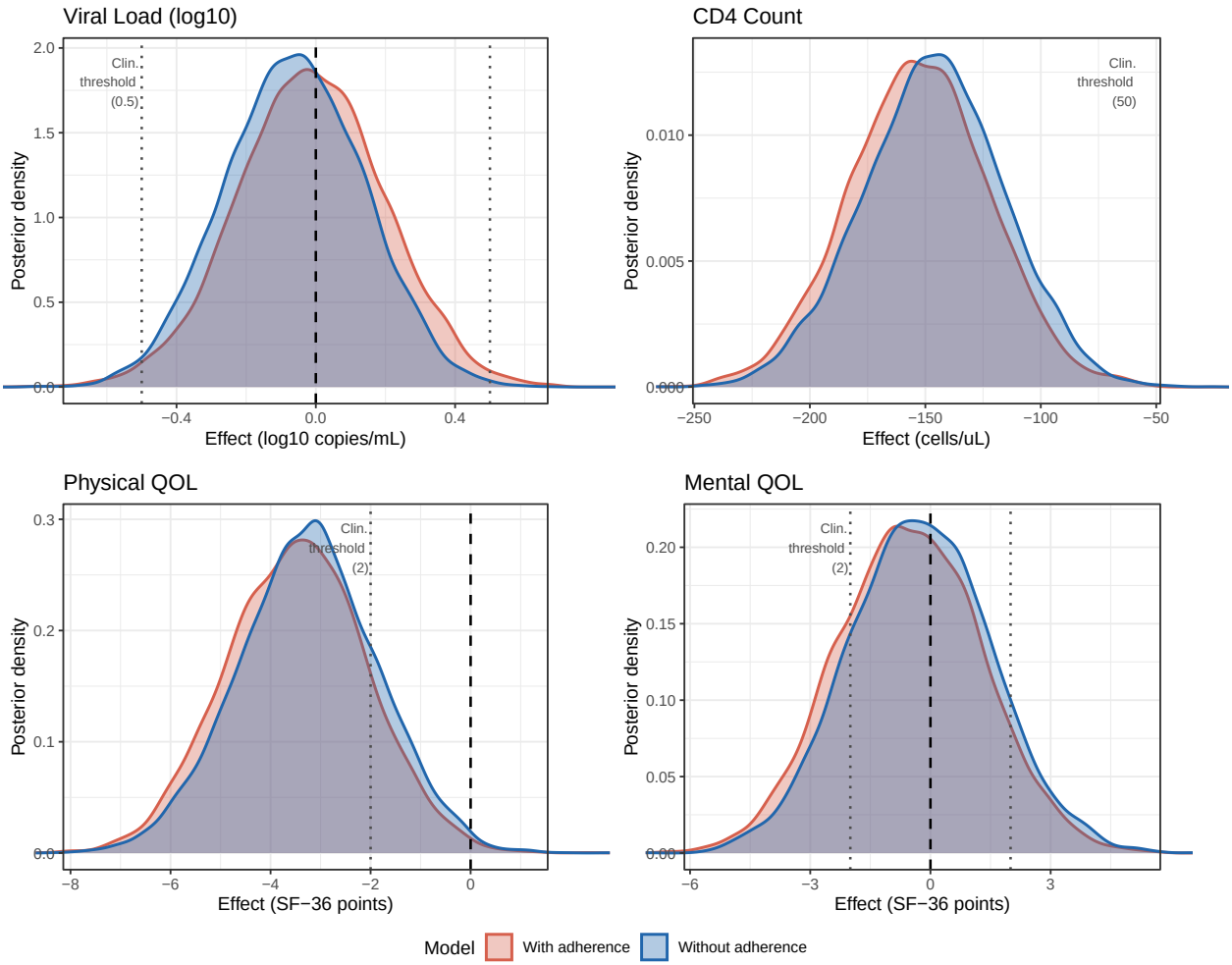

Figure 2: Posterior distributions of the hard drug use coefficient by outcome. Dashed line = no effect; dotted lines = clinical thresholds.

Reproducible Research Information

GitHub Repository: https://github.com/mhaque-99/BIOS6624/tree/main

File name for Code: Project1/Code/Project 1 Report.rmd

File name for Final report:Project1/Reports/Project 1 Report.pdf

Directory for the raw dataset: C:/Users/mahim/OneDrive - The University of Colorado Denver/BIOS6624/Project1/DataRaw

Directory for the cleaned/analytic dataset: C:/Users/mahim/OneDrive - The University of Colorado Denver/BIOS6624/Project1/DataProcessed

Supplimentary Tables and Figures

Table 5: MCMC convergence diagnostics for all Bayesian models.

Outcome	Model	Max R-hat	Min Bulk ESS	Min Tail ESS	Converged?
Viral Load (log10)	No adherence	1.002	3539	4815	Yes
CD4 Count	No adherence	1.002	3797	5204	Yes
Physical QOL	No adherence	1.002	3711	5080	Yes
Mental QOL	No adherence	1.002	3719	4721	Yes
Viral Load (log10)	With adherence	1.001	3426	4801	Yes
CD4 Count	With adherence	1.002	3608	4884	Yes
Physical QOL	With adherence	1.002	3516	5173	Yes
Mental QOL	With adherence	1.003	3600	4426	Yes

Table 6: LOO-IC model fit statistics for all Bayesian models.

Outcome	Model	ELPD-LOO	LOO-IC	p_LOO
Viral Load (log10)	No adherence	-720.8	1441.6	10.3
CD4 Count	No adherence	-3063.1	6126.2	12.0
Physical QOL	No adherence	-1610.9	3221.9	11.7
Mental QOL	No adherence	-1723.0	3446.1	11.6
Viral Load (log10)	With adherence	-719.2	1438.4	11.9
CD4 Count	With adherence	-3061.8	6123.6	12.6
Physical QOL	With adherence	-1610.1	3220.2	12.3
Mental QOL	With adherence	-1723.1	3446.3	12.9

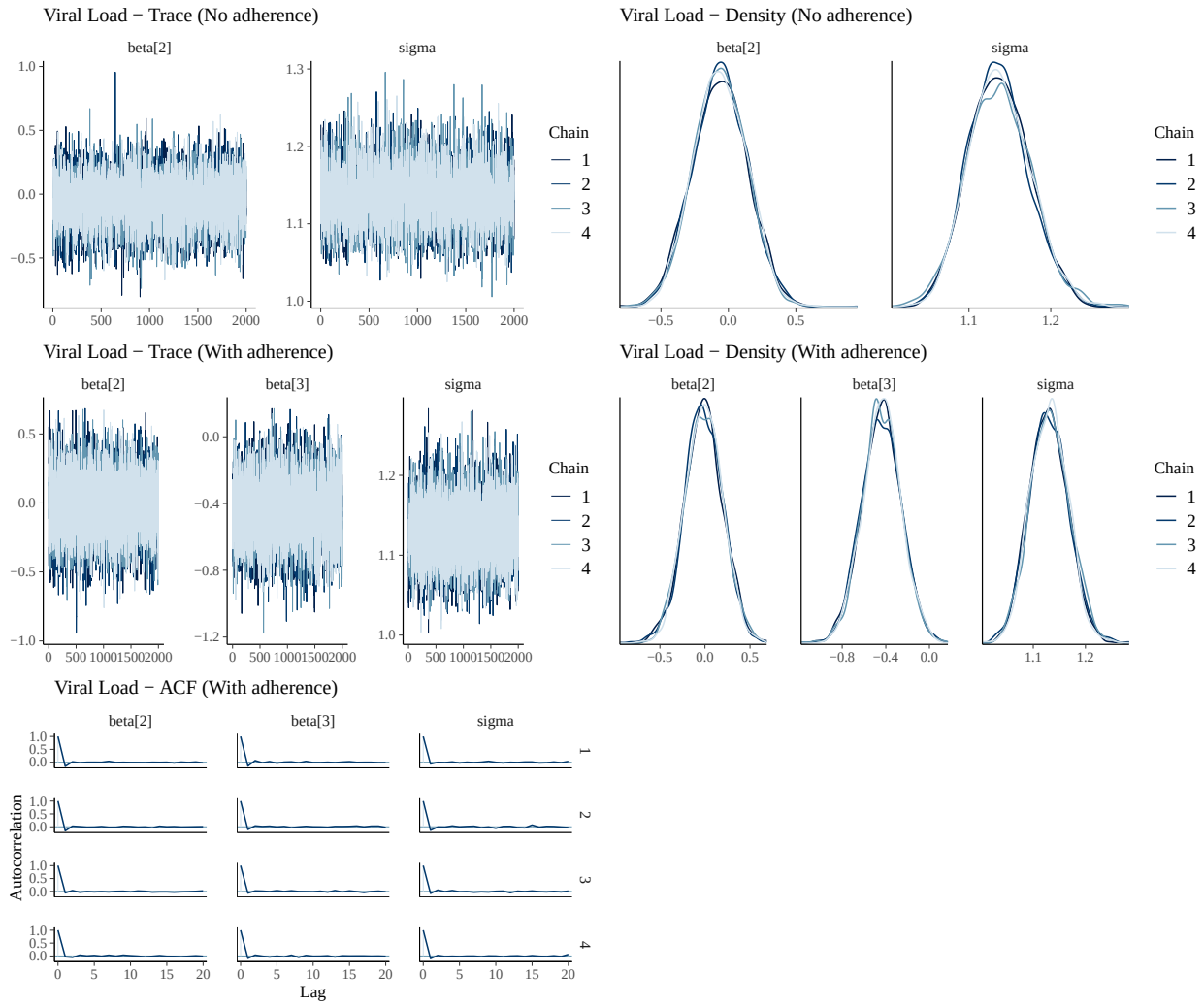

Figure 3: MCMC diagnostics for Viral Load model. Top row: unadjusted model. Bottom rows: adherence-adjusted model.

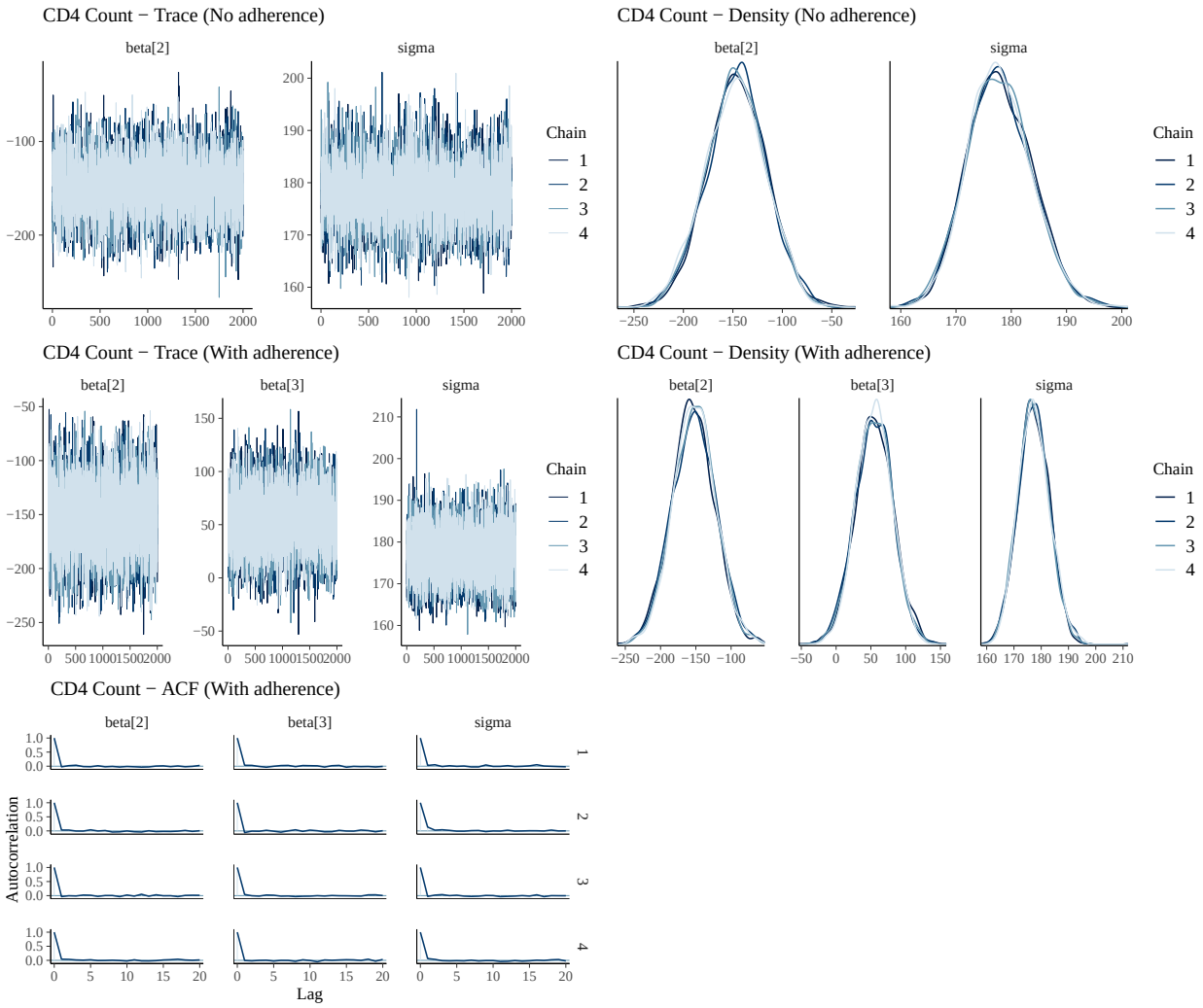

Figure 4: MCMC diagnostics for CD4 Count model.

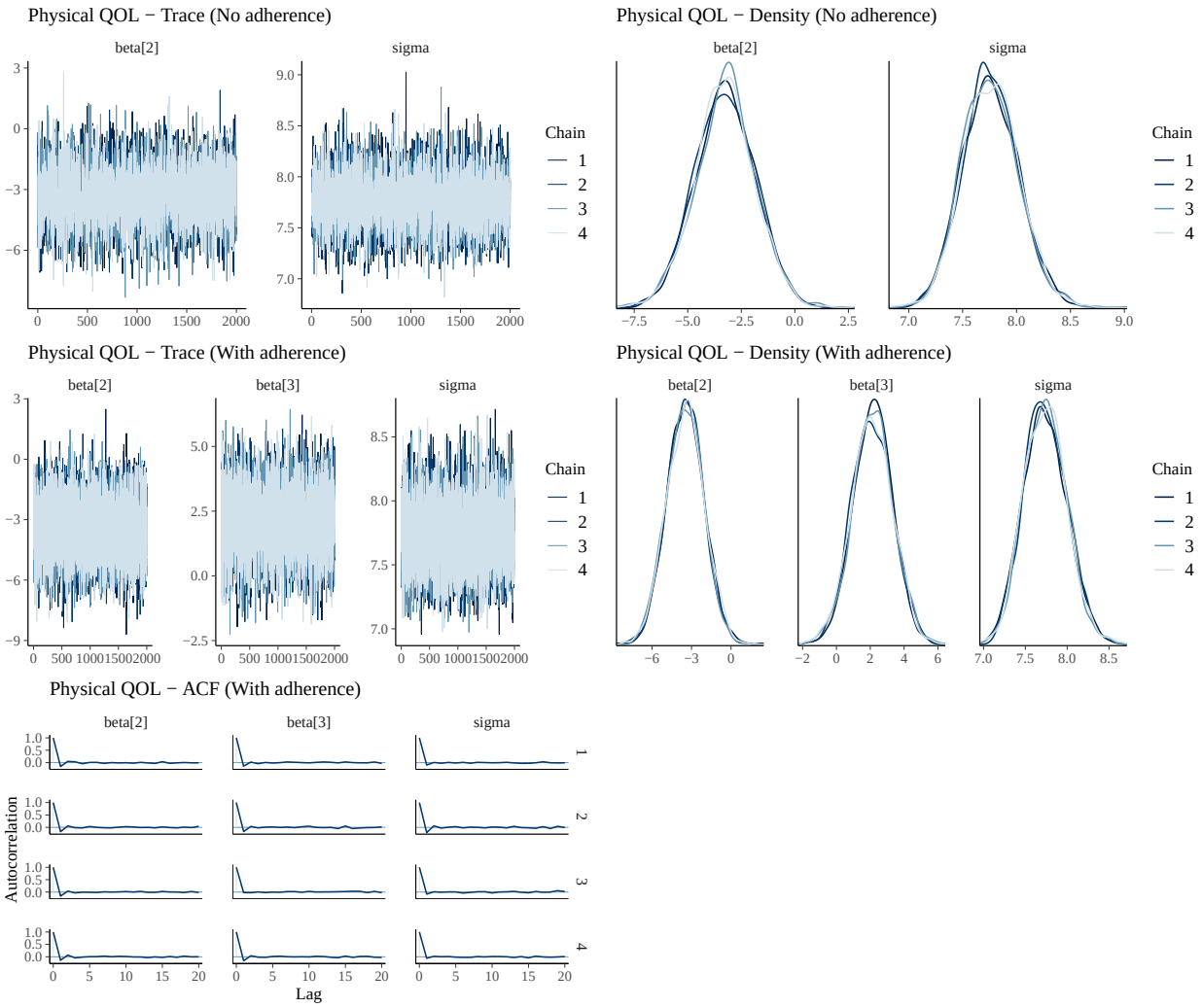

Figure 5: MCMC diagnostics for Physical QOL model.

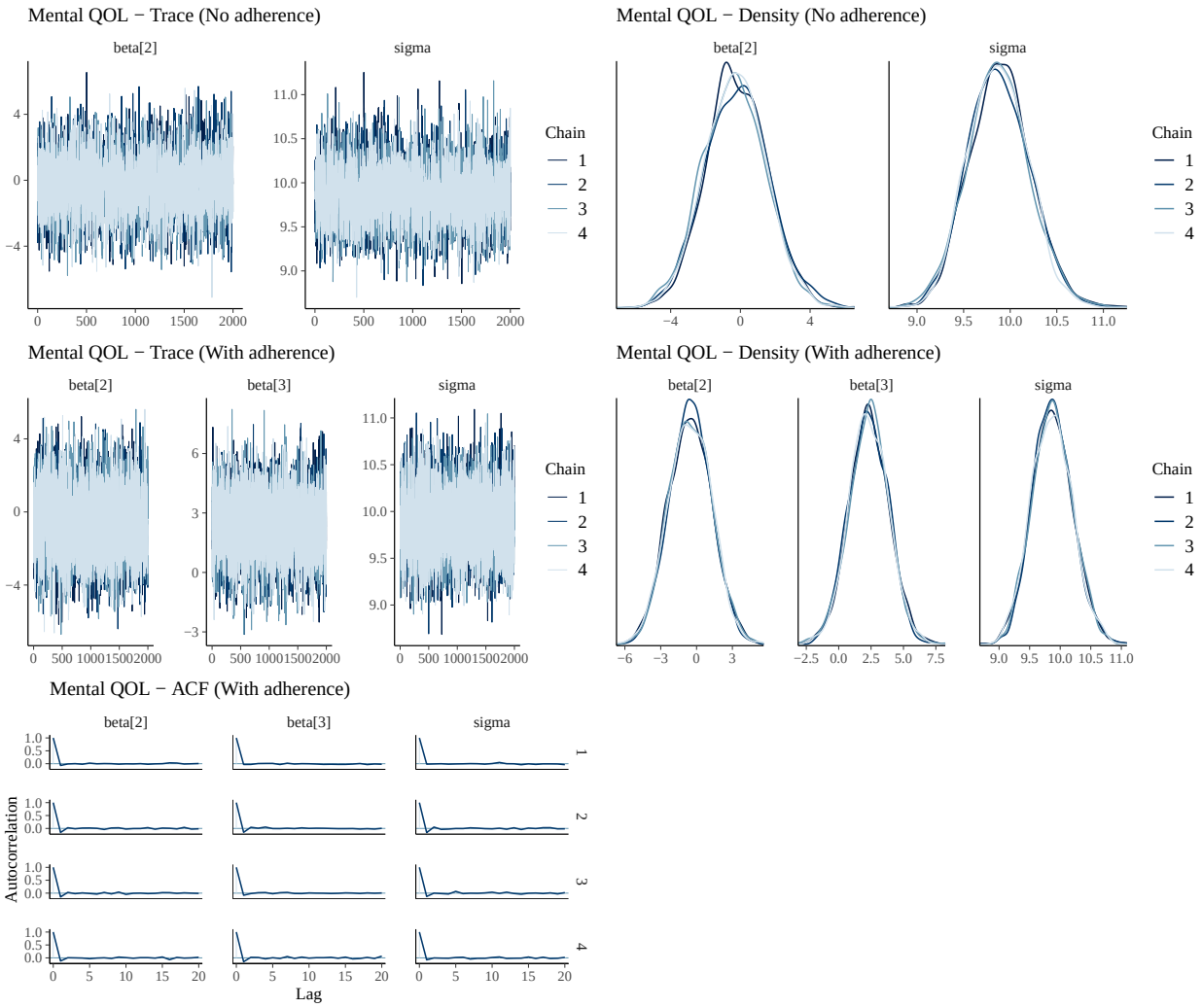

Figure 6: MCMC diagnostics for Mental QOL model.